Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +91 8591441019 | www.github.com/Arishtanemi4 |
www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

AI / GenAI Engineer | Forward Deployed Engineer | Machine Learning Engineer | Data Scientist | Full Stack Developer | Data Engineer | MLOps Engineer

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

PROFILE SUMMARY

AI Solutions Engineer and Forward Deployed Engineer with **3 years** of industry experience building and deploying production **GenAI applications** end to end: **RAG**, **Azure OpenAI**, **LlamaIndex**, **pgvector**, **FastAPI**, **ReactJS** and **AWS**. Hands-on across **SQL data engineering**, **Python** analytics, **Power BI** and **MLOps** (**Docker**, **Jenkins/GitLab CI/CD**), and a de facto team lead for up to 4 interns. Recently completed an **MSc in Data Science at the University of Bristol (UK)**, awaiting result. Seeking GenAI, ML, Data Science, Full Stack and Data Engineering roles in India.

KEY SKILLS

- **AI / ML:** Generative AI, RAG, LLM applications, Azure OpenAI, LlamaIndex, Llama-3, embeddings, vector databases (pgvector), agentic tool calling, prompt engineering, scikit-learn, CNN.
- **Languages:** Python (pandas, FastAPI, uvicorn, scikit-learn, fuzzywuzzy), R (dplyr, tidyverse, ggplot2), SQL, NodeJS (Express, ReactJS), Shell.
- **Cloud / DevOps:** AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, Textract, Security Groups, CloudFormation templates), Docker, Jenkins, GitLab CI/CD, Nginx, Linux, LDAP (RBAC).
- **Databases and Tools:** PostgreSQL (schema design, functions and procedures, pgvector), SQLite, Power BI, Jupyter, Anaconda, Git, VS Code, Claude Code.

WORK EXPERIENCE

HDFC Asset Management Company Ltd. **July 2022 – July 2025**

AI Solutions Engineer / Forward Deployed Engineer

GenAI / Retrieval Augmented Generation (RAG) Engineer

- Designed and built **GenAI applications** end to end (**ReactJS** frontend, **FastAPI** backend, **RAG** logic, **vector database** and **embeddings** pipeline) using **LlamaIndex**, **Azure OpenAI GPT-3.5** and **Llama-3** to answer department-specific queries for Compliance, Client Service and Investment teams.
- Maintained department-wise **vector databases** (**pgvector**, **Azure OpenAI embeddings**) for **semantic search**; allocated models by cost, with **Llama-3 self-hosted on a GPU EC2 instance** for Client Service and **Azure OpenAI** for other departments.
- Built **AI agents** that call **Python functions as tools** to run fund-comparison calculations and plot them into **PDF** reports on user request.
- Developed a **Python** automation job (**Azure OpenAI**, earlier **AWS Textract**, with **fuzzywuzzy** matching) on **AWS SageMaker**, scheduled via **EventBridge** and **Lambda**, to parse **300+ daily emails** from credit rating agencies into a daily consolidated analyst report, **saving 60 manual analyst hours a month**.

Full Stack and Forward Deployed Engineering

- Worked directly with Client Service, Compliance and Investment stakeholders, building to their requirements; the only engineer alongside the manager in year 1, later a **de facto team lead** guiding **2–4 interns**.
- Served the **ReactJS** frontend with **Nginx** on one **AWS EC2** instance and the **FastAPI microservices** (**Docker**, **uvicorn**) on another, with **RBAC** through **LDAP** authentication.
- Recreated the **Power BI** dashboards in **ReactJS**, making the data available across the organization at lower cost than scaling Power BI licences.

Data Analyst

- Analyzed customer data and drew insights using **Python Pandas**.
- Developed **Power BI** dashboards for the investment team's fortnightly credit report analysis, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**.
- Identified customer buying patterns using **Market Basket Analysis** (**Apriori Algorithm**) for the marketing team.

Data Engineer

- Designed the entire **Normalized SQL Schema** (**PostgreSQL**) and developed **Optimized SQL** queries, **SQL Functions** and **Procedures** for preprocessing and efficient data insertion.
- Wrote **Python** ingestion scripts for the enterprise-licensed **Morningstar API**, refreshing data fortnightly (fixed income funds) and monthly (equity funds) for every fund in the Indian mutual fund market (2023–2025), and integrated it with legacy data held in analysts' Excel files.

MLOps / DevOps / Cloud Infrastructure

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** utilizing **EC2, S3, ECR and Lambda**, provisioning EC2 instances with security-team **CloudFormation** templates.
- Restricted app access to company private IPs with **AWS Security Groups** and deployed an **Nginx reverse proxy**, one of the fixes that took the app from failing to passing the security team's **VAPT**.
- Built the deployment automation with **Jenkins** and **GitLab** (**CI/CD** pipeline): a one-click deploy of the dev or prod branch for separate frontend and backend repositories, each in its own **Docker** container.

PROJECTS

- **University of Bristol MSc Dissertation with AstraZeneca:** CellLineSelector (team project; algorithm to rank best suited cancer cell lines based on target inclusion and exclusion genes. Personal focus: data preprocessing, scoring algorithm and RAG layer.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app (108 AI personas across 12 archetypes, profile evaluation, vision-model photo scoring. React, FastAPI.) | GitHub: https://github.com/Arishtanemi4/wingman.git
- **MSc Applied Data Science Group Project:** VisionCaptioner (team project; CNN/ViT/ResNet encoders vs LSTM/attention-based decoders for image captioning, benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator (car trained using a CNN model).

ACHIEVEMENTS

- **HDFC AMC Best Team Project 2024 and 2025 (internal recognition):** rating-change email automation (2024) and department-specific RAG chatbot (2025).
- **Project Hack 27:** Rolls Royce Derby 4th Place winner. (Human centric data analytics tool.)
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Spec-driven AI dating app.)
- **University of Bristol Datathon (Lloyd's Bank):** 3rd Place winner. (Credit card fraud detection.)

EDUCATION

University of Bristol, UK – MSc. Data Science (awaiting result) — **September 2025 – September 2026**
NMIMS, Mumbai – B.Tech IT (Dual Degree, CGPA 3.27/4) — **June 2017 – April 2022**
NMIMS, Mumbai – MBA Technology Management (Dual Degree) — **June 2017 – April 2022**

CERTIFICATIONS

- **Udemy:** Practical Deep Learning (Intel); Applied Data Science with Python; Mathematical Foundation for ML and AI.
- **Claude Code in Action:** https://verify.skilljar.com/c/4rride9vnwmv